

AMENDMENT NO. 2 JULY 2024
TO
IS 17483 (PART 1) : 2020 GEOSYNTHETICS — GEOCELLS — SPECIFICATION
PART 1 LOAD BEARING APPLICATION

(Foreword, Para 6) — Substitute the following for the existing:

‘The composition of the Committee responsible for the formulation of this standard is given in Annex H.’

(Page 1, clause 3.2, see also Amendment No. 1) — Substitute the following for the existing:

‘3.2 Style Number — A theoretical numerical identifier assigned to geocells, indicating their distinct design characteristics in terms of weld spacing (mm) and cell depth (mm). For instance, ‘a × b’ type geocell refers a geocell having weld spacing of ‘a’ mm and a cell depth of ‘b’ mm.’

(Page 1, clause 5.2, see also Amendment No. 1) — Substitute the following for the existing:

‘5.2 Geocells used for load bearing application shall have a depth ranging from 125 mm to 200 mm (*see* Note) and weld spacing ranging from 300 mm to 445 mm. The expanded individual geocell and geocell panel are shown in Fig. 2 (a) and Fig. 2 (b) respectively for illustration. The following tolerance shall be permissible for depth and weld spacing:

<i>Sl No.</i>	<i>Dimension</i>	<i>Tolerance, Percent</i>
(1)	(2)	(3)
i)	Depth	$\left\{ \begin{array}{c} +3 \\ -2 \end{array} \right\}$
ii)	Weld spacing	$\left\{ \begin{array}{c} +3 \\ -3 \end{array} \right\}$

NOTE — In specific cases such as infilling with concrete or self-compacting material, geocells of depth above 200 mm and up to 300 mm may also be used.’

(Page 2, clause 5.2.1) — Substitute the following for the existing:

‘For effective confinement h/D_{eq} ratio shall be from 0.68 to 1.25. Equivalent diameter of the geocell pocket (D_{eq}) shall be measured by the formula as follows:

$$D_{eq} = \frac{1}{2} \sqrt{d_1^2 + d_2^2}$$

where

h = depth of the geocell;

D_{eq} = equivalent diameter of the geocell pocket;

d_1 = length of the geocell pocket; and

d_2 = width of the geocell pocket.

The geocell pocket size [length (d_1) × width (d_2)] shall be measured in accordance with the method given in Annex G.’

[Page 1, Fig. 2 (c), see also Amendment No. 1] — Delete.

(Page 14, Annex G) — Insert the following new Annex G and renumber existing Annex G to Annex H:

‘ANNEX G

(Clause 5.2.1)

METHOD FOR DETERMINATION OF GEOCELL POCKET SIZE

G-1 ATMOSPHERIC CONDITION FOR TESTING AND CONDITIONING OF TEST SPECIMENS

The test shall be carried out in the standard atmosphere of (65 ± 2) percent relative humidity and $27\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$ temperature (*see* IS 196).

G-2 TESTING EQUIPMENT

Measuring tape or ruler, metallic frames with protruding bars at uniform spacing to hold and extend the geocell panel.

G-3 TEST SPECIMEN

Take a geocell panel having minimum 29 cells and 10 cells along its length and width respectively.

G-4 PROCEDURE

G-4.1 Extend the geocell panel across the metallic frame to attain the optimal opening without giving any bowing along the edges and ensure a flat and uniform configuration, maintaining the integrity of the geocell structure (*see* Fig. 15).

G-4.2 Measure the length (l) of the geocell panel by using measuring tape or ruler which spans from one wall of geocell to other (*see* Fig. 16). Similarly, measure the width (w) of the geocell panel ensuring the measurement spans from one weld point to another at both ends of the panel (*see* Fig. 16). Ensure accuracy by aligning the measuring tape or ruler parallel to the axis of the geocell panel, capturing the full extent of geocell panel along its length and width.

G-4.3 Determine the number of cells in lengthwise and widthwise directions in the geocell panel.

G-5 CALCULATION

Calculate the length and width of geocell pocket by dividing length (l) and width (w) of the geocell panel by the number of cells along its length and width respectively using the following formulas:

$$d_1 = \frac{l}{n_l}$$

and

$$d_2 = \frac{w}{n_w}$$

where

d_1 = length of the geocell pocket;

l = length of geocell panel;

n_l = number of cells in lengthwise direction;

d_2 = width of the geocell pocket;

w = width of geocell panel; and

n_w = number of cells in lengthwise direction.

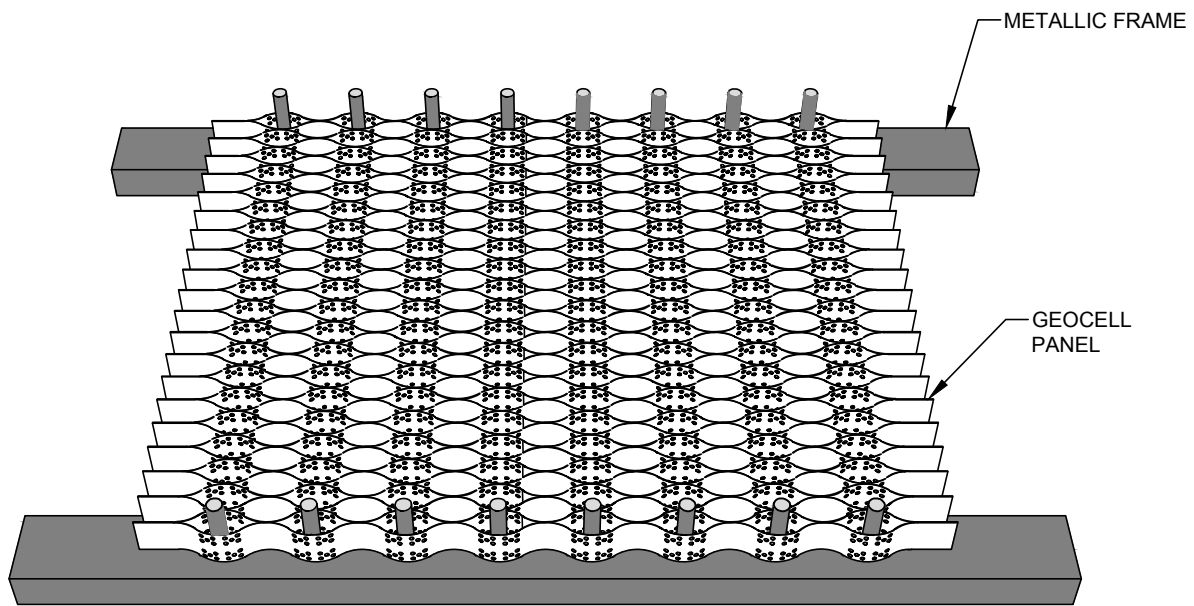


FIG. 15 PLACEMENT OF GEOCELL PANEL ON METALLIC FRAME

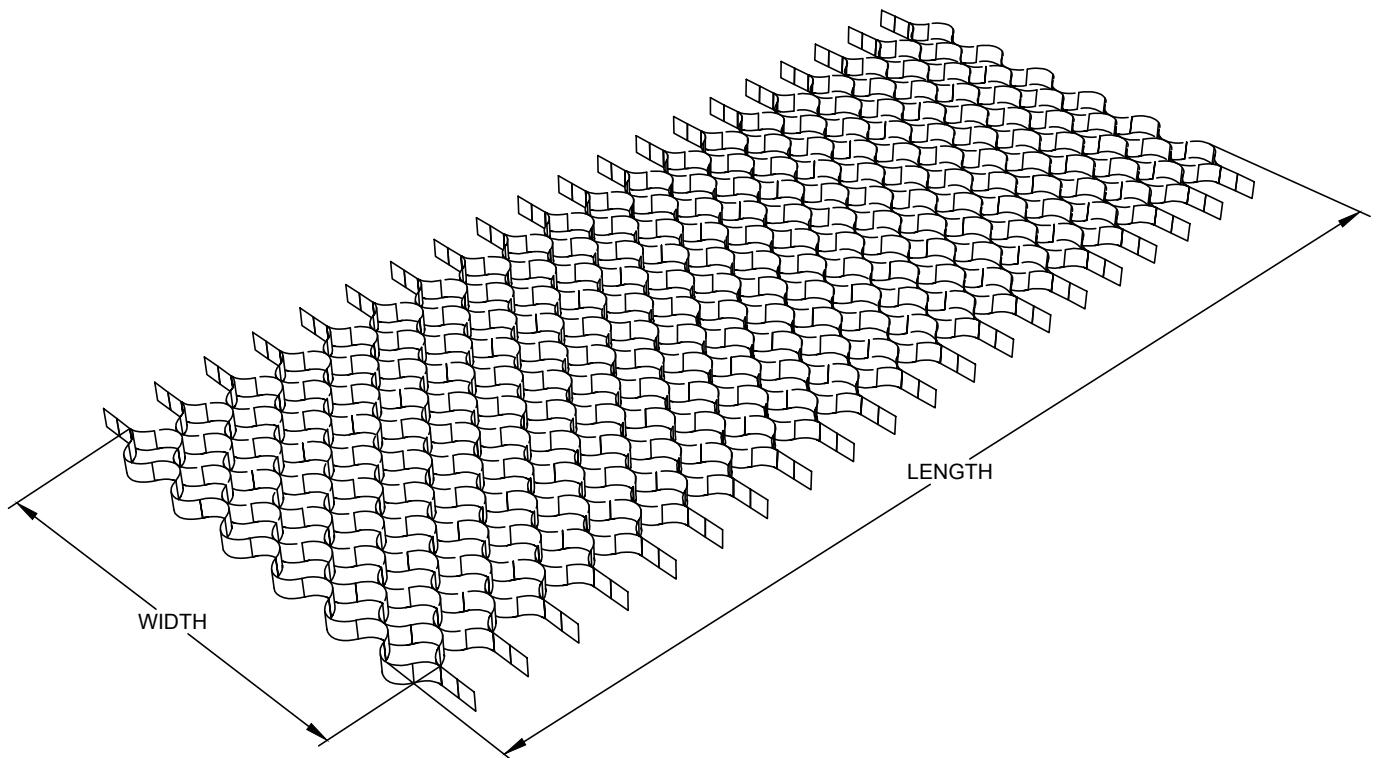


FIG. 16 EXPANDED GEOCELL PANEL